

A CHARACTERISTIC-TWO COUNTEREXAMPLE TO THE SHALEV–SMOKTUNOWICZ CONJECTURE

ABSTRACT. We construct, for every $r \geq 1$, a left brace A_r of order 2^{8r+1} whose additive group is elementary abelian and whose multiplicative group is a powerful 2-group of nilpotency class two, while A_r is not right nilpotent. The case $r = 1$ has order 512 and multiplicative group G satisfying $G' = G^4 \cong C_2^2$ and $\exp(G) = 8$. This gives a characteristic-two counterexample to the Shalev–Smoktunowicz conjecture and answers Question 10.2 of Verbeke. Together with Verbeke’s odd-prime examples, it completes the negative answer for every prime. The construction is a characteristic-two adaptation of Verbeke’s kernel-graph method; the essential modification is a truncation degree divisible by four.

1. INTRODUCTION

For a left brace A , write

$$\lambda_a(b) = -a + a \circ b, \quad a * b = \lambda_a(b) - b.$$

For additive subgroups $X, Y \leq (A, +)$, let $X * Y$ denote the additive subgroup generated by the elements $x * y$. The left and right series are

$$A^1 = A, \quad A^{n+1} = A * A^n, \quad A^{(1)} = A, \quad A^{(n+1)} = A^{(n)} * A.$$

A finite 2-group G is powerful when $G' \leq G^4 = \langle g^4 : g \in G \rangle$ [1].

Shalev and Smoktunowicz asked whether a brace of prime-power order with powerful adjoint group must be strongly nilpotent [2, Question 1]. Vendramin recorded the corresponding right-nilpotence statement as Problem 5.15 [3, Problem 5.15]. Following Verbeke, we refer to this right-nilpotence statement as the Shalev–Smoktunowicz conjecture. Verbeke disproved it for every odd prime and asked whether an analogous counterexample exists for $p = 2$ [4, Question 10.2]. Together with his odd-prime examples, Theorem 1 settles the conjecture negatively for every prime.

Our construction is the characteristic-two instance of Verbeke’s kernel-graph mechanism [4, Proposition 2.2]. The new point is to decouple the truncation degree from the characteristic and take it divisible by four; this places the commutators inside fourth powers, as required for a powerful 2-group.

Theorem 1. *For every $r \geq 1$ there exists a left brace A_r such that*

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- (1) $(A_r, +) \cong C_2^{8r+1}$;
- (2) if $G_r = (A_r, \circ)$, then G_r has nilpotency class two, $G'_r \cong C_2^2$, and $G'_r \leq G_r^4$;
- (3) A_r is left nilpotent, is not right nilpotent, and $\text{Soc}(A_r) = 0$.

For $r = 1$, one has

$$|A_1| = 512, \quad G'_1 = G_1^4 \cong C_2^2, \quad \exp(G_1) = 8.$$

Corollary 2. *The order- 3^7 example in [4] is not minimal among finite left braces with powerful multiplicative group that are not right nilpotent.*

Proof. Theorem 1 gives an example of order $2^9 = 512 < 3^7 = 2187$. $\square$

No claim is made that 512 is the smallest possible order.

2. THE CONSTRUCTION

Fix $r \geq 1$, put $m = 4r$, and work over $\mathbb{F}_2$. Let

$$B = \mathbb{F}_2[x, y]/(xy, x^{m+1} + y^{m+1}), \quad J = (x, y), \quad z = x^{m+1} = y^{m+1}.$$

Lemma 3. *The ideal J has basis*

$$x, x^2, \dots, x^m, y, y^2, \dots, y^m, z.$$

In particular, $\dim_{\mathbb{F}_2} J = 2m + 1$ and $J^{m+1} = \mathbb{F}_2 z$, $J^{m+2} = 0$. The assignments

$$D(x) = y^m, \quad D(y) = x^m$$

define a derivation D of B . If

$$W = D(J) = \text{span}_{\mathbb{F}_2}\{x^m, y^m\},$$

then $D^2 = 0$, $W^2 = 0$, and $\phi = 1 + D$ is an algebra automorphism of order two.

Proof. With lexicographic order $x > y$, the polynomials

$$xy, \quad x^{m+1} + y^{m+1}, \quad y^{m+2}$$

form a Gröbner basis of the defining ideal. The standard monomials are $1, x, \dots, x^m, y, \dots, y^{m+1}$, which proves the basis assertion and the statements about the powers of J .

The derivation preserves the relations because

$$D(xy) = y^{m+1} + x^{m+1} = 0, \quad D(x^{m+1} + y^{m+1}) = 0.$$

Moreover, $D(x^i) = D(y^i) = 0$ for every $i \geq 2$. Hence $D^2 = 0$ and $D(J) = W$, while $W^2 = 0$. Therefore

$$(a + D(a))(b + D(b)) = ab + D(ab)$$

for all $a, b \in B$, so $\phi = 1 + D$ is multiplicative. Since $\phi^2 = 1$ and $D \neq 0$, it has order two. $\square$

Let $U = 1 + J \leq B^\times$.

Lemma 4. *There is a homomorphism $\ell : U \rightarrow \mathbb{F}_2$ such that*

$$\ell(1+x) = \ell(1+y) = 0, \quad \ell(1+z) = 1, \quad \ell \circ \phi = \ell.$$

Proof. Since U is abelian, U/U^2 is an $\mathbb{F}_2$ -vector space. The classes of $1+x$, $1+y$, and $1+z$ are linearly independent. Indeed,

$$(1+x)^\alpha(1+y)^\beta(1+z)^\gamma = 1 + \alpha x + \beta y + \gamma z,$$

whereas every element of U^2 is a square $1+a^2$ and has no x -, y -, or z -component; the last assertion uses that $m+1$ is odd. A linear functional on U/U^2 with the prescribed values therefore lifts to a homomorphism $\ell : U \rightarrow \mathbb{F}_2$.

For $a \in J$, write $a = \alpha x + \beta y + a_2$, where a_2 has no linear terms. Then $D(a_2) = 0$, $a_2 W = 0$, and

$$D(a) = \alpha y^m + \beta x^m, \quad aD(a) = \alpha\beta(x^{m+1} + y^{m+1}) = 0.$$

Thus, for $u = 1+a$,

$$\phi(u) = u(1 + D(a)).$$

Because $m = 4r$,

$$1 + D(a) = (1 + \alpha y^r + \beta x^r)^4 \in U^4 \subseteq U^2.$$

Every homomorphism from U to $\mathbb{F}_2$ vanishes on U^2 , so $\ell(\phi(u)) = \ell(u)$. $\square$

Put $\varepsilon(a) = \ell(1+a)$ and define, for $a, b \in J$,

$$(1) \quad \lambda_a(b) = (1+a)\phi^{\varepsilon(a)}(b), \quad a \circ b = a + \lambda_a(b).$$

Proposition 5. *The operations $+$ and $\circ$ make J a left brace, denoted A_r . Its star product is*

$$(2) \quad a * b = ab + \varepsilon(a)D(b) + \varepsilon(a)aD(b).$$

Proof. This is the kernel-graph construction of [4, Proposition 2.2]; we include the short verification. The map

$$\chi : U \rtimes \langle \phi \rangle \longrightarrow \mathbb{F}_2, \quad \chi(u\phi^t) = \ell(u) + t,$$

is a homomorphism by Lemma 4. Its kernel is

$$H = \{u\phi^{\ell(u)} : u \in U\}.$$

Acting on U by

$$(u\phi^t) \cdot v = u\phi^t(v),$$

the group H is regular because the element indexed by u sends 1 to u . Under the identification $1+a \leftrightarrow a$, its affine action on J is

$$b \longmapsto a + (1+a)\phi^{\varepsilon(a)}(b).$$

The standard regular-subgroup construction therefore gives the left-brace operation (1). Since $\phi^{\varepsilon(a)} = 1 + \varepsilon(a)D$, expansion gives (2). $\square$

3. PROOF OF THE THEOREM

Let $A = A_r$ and $G = (A, \circ)$, identified with the kernel graph above. Lemma 3 gives $(A, +) \cong C_2^{2m+1} = C_2^{8r+1}$. Put $K = \ker \ell$. Since $\ell(1+z) = 1$, the element $c = (1+z)\phi$ belongs to G , and

$$G = K\langle c \rangle.$$

Conjugation by c induces ϕ on the abelian group K . Hence

$$G' = \langle k^{-1}\phi(k) : k \in K \rangle.$$

For $k = 1 + a$, Lemma 4 gives $k^{-1}\phi(k) = 1 + D(a)$, so $G' \leq 1 + W$. Since $1 + x, 1 + y \in K$,

$$(1+x)^{-1}\phi(1+x) = 1 + y^m, \quad (1+y)^{-1}\phi(1+y) = 1 + x^m.$$

Therefore

$$(3) \quad G' = 1 + W \cong C_2^2.$$

The subgroup $1 + W$ is fixed pointwise by ϕ and is central in U , hence central in G . Thus G has class at most two; by (3), its class is exactly two.

Let $g = u\phi^t \in G$, where $u = 1 + a$ and $t = \ell(u)$. Then

$$g^2 = u^2(1 + D(a))^t, \quad g^4 = u^4.$$

Consequently,

$$(4) \quad \begin{aligned} G^4 &= U^4 = 1 + \text{im}(a \mapsto a^4) \\ &= 1 + \text{span}_{\mathbb{F}_2}\{x^4, x^8, \dots, x^m, y^4, y^8, \dots, y^m\}. \end{aligned}$$

In particular, $G' = 1 + W \leq G^4$, so G is powerful. If $r = 1$, then (4) gives $G^4 = 1 + W = G'$. Moreover, every fourth power then lies in the exponent-two group $1 + W$, so $\exp(G) \leq 8$; the element $1 + x \in G$ (which lies in K) has order eight, since $(1+x)^4 = 1 + x^4 \neq 1$ and $(1+x)^8 = 1$. Hence $\exp(G) = 8$.

We next examine the right series. Let

$$T = \text{span}_{\mathbb{F}_2}\{x^m, y^m, z\}.$$

Since $1 + x^m, 1 + y^m \in U^4$, one has $\varepsilon(x^m) = \varepsilon(y^m) = 0$, while $\varepsilon(z) = 1$. Formula (2) gives

$$x^m * x = z, \quad y^m * y = z, \quad z * x = y^m, \quad z * y = x^m.$$

Also $TJ \subseteq \mathbb{F}_2 z$, $D(J) = W$, and $TW = 0$, so $T * A \subseteq T$. The displayed products show the reverse inclusion and also show $T \subseteq A^{(2)}$. Thus

$$T * A = T, \quad T \subseteq A^{(n)} \quad (n \geq 2),$$

and A is not right nilpotent.

For the left series, (2) gives $A * A \subseteq J^2$. Since $\varepsilon(x) = \varepsilon(y) = 0$,

$$x * x^i = x^{i+1}, \quad y * y^i = y^{i+1} \quad (1 \leq i \leq m),$$

so $A * A = J^2$. Moreover, $D(J^n) = 0$ for $n \geq 2$, and therefore $A * J^n = J^{n+1}$. Inductively,

$$A^n = J^n, \quad A^{m+1} = \mathbb{F}_2 z, \quad A^{m+2} = 0.$$

Hence A is left nilpotent.

Finally, suppose $a \in \text{Soc}(A) = \ker \lambda$ and put $e = \varepsilon(a)$. From (1),

$$\lambda_a(x) = x + ey^m + ax + eay^m.$$

Neither ax nor ay^m has a y^m -component, so $\lambda_a(x) = x$ implies $e = 0$. It follows that $(1 + a)b = b$ for all $b \in J$, hence $aJ = 0$. The basis in Lemma 3 gives $\text{Ann}_J(J) = \mathbb{F}_2 z$, so $a = \gamma z$. But then $0 = e = \varepsilon(a) = \gamma$, and therefore $a = 0$. This proves $\text{Soc}(A) = 0$ and completes the proof of Theorem 1.

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